

CHAPTER 3: ROUTING FUNDAMENTALS

Routing is the process of selecting paths in a network along which to send data. A router is essentially a "Post Office" for the internet. Its primary job is to maintain a **Routing Table** to know where different networks are located.

STATIC ROUTING

- A manual process where a network administrator explicitly tells the router, "To get to Network X, go through Router Y." The router doesn't "think" for itself; it only follows your specific instructions.
- In Static Routing, a network administrator **manually** types the paths into the router's configuration.
- **Pros:** Secure (no protocols talking to each other), uses zero CPU/bandwidth, and is predictable.
- **Cons:** Human error is common, and it doesn't scale. If a link goes down, the router doesn't "know" how to find a new way unless you manually change it.
- **The Command Syntax:** `ip route [destination_network] [mask] [next_hop_address]`
- **Standard Static Routes** – Mapping a specific destination network to a specific exit path.
- **Default Static Routes** – A catch-all route used when the router doesn't have a specific match in its table.

TYPES OF STATIC ROUTING

- **RECURSIVE LOOKUPS** – In most static route configurations, you point toward a **Next-Hop IP address** rather than an exit interface. Because the router doesn't "know" where that Next-Hop IP is physically located, it has to perform two searches in its routing table to figure out where to send the packet.
 - **The Example:** Imagine your router (R1) has a neighbor (R2) at IP 10.0.0.2. You want to reach a remote network 192.168.2.0.
 - **You configure:** `ip route 192.168.2.0 255.255.255.0 10.0.0.2`
 - **The Packet arrives:** Destination 192.168.2.50.
 - **Lookup #1:** The router looks at the table and finds the route. It says: "To get there, I must go to 10.0.0.2." (But it doesn't know which port leads to 10.0.0.2 yet!)
 - **Lookup #2:** The router looks at the table again to find 10.0.0.2. It finds a "Connected" route saying 10.0.0.0/30 is on GigabitEthernet0/0.
- **FLOATING STATIC ROUTES** – A static route with a higher "Administrative Distance" (AD) than the primary route. It stays "invisible" unless the main link fails.
 - **Default Static AD:** 1 (Very trusted)
 - **Floating Static AD:** Anything higher than the primary (e.g., 10, 50, or 250)
 - **The Example:** You have a fast **Fiber** link (Primary) and a slow **Satellite** link (Backup).
 - **Primary Route:** `ip route 172.16.1.0 255.255.255.0 10.1.1.2 (AD is 1)`
 - **Backup Route:** `ip route 172.16.1.0 255.255.255.0 10.2.2.2 250 (AD is 250)`

INTRODUCTION TO DYNAMIC ROUTING

- In Dynamic Routing, routers use **protocols** (like OSPF or EIGRP) to talk to each other. They "gossip" about the networks they know.
- **Pros:** If a link dies, the routers automatically calculate a new path. It scales easily to thousands of routers.

- **Cons:** Uses some CPU and bandwidth to send "Hello" packets; can be more complex to secure.
- **Interior Gateway Protocols (IGP)** – Protocols used to route data within a single organization (like a university or a company).
 - **Example: OSPF (Open Shortest Path First).** It looks for the "fastest" path, not just the shortest.
- **Metrics and Administrative Distance (AD)** – How a router decides which path is "better" when it hears about a destination from two different sources.
- **CONVERGENCE TIME**
 - The time it takes for all routers in a network to agree on the current map after a change (like a cable being unplugged).
 - High-speed protocols like **OSPF** or **EIGRP** converge in seconds. Old protocols like **RIP** can take minutes.
- **METRICS**
 - The "Cost" assigned to a path. If there are two ways to get to a destination, the router chooses the one with the lowest metric.
 - **Metric Types:**
 - **Hop Count (Used by RIP):** Counts how many routers are in the way.
 - **Bandwidth (Used by OSPF):** Chooses the fastest cable.

ROUTING TABLE ANALYSIS

To be a good network engineer, you must be able to read a routing table. On a Cisco router, you use the command `show ip route`.

Code	Meaning	Description
C	Connected	A network physically plugged into the router.
L	Local	The specific IP address assigned to the router's interface.
S	Static	A path manually entered by an admin.
O	OSPF	A path learned dynamically via the OSPF protocol.
D	EIGRP	A path learned dynamically via the EIGRP protocol.

Routing Table Analysis – The "brain" of the router. It is a list of all known paths stored in RAM.

- **Interpreting Route Codes** – The letters at the start of a route entry that tell you where the info came from.
 - **Example:**
 - **C (Connected):** Directly plugged into the router.
 - **S (Static):** Manually typed by an admin.
 - **O (OSPF):** Learned dynamically through OSPF.

The routing table is not just a list; it is a prioritized hierarchy. When a packet arrives, the router follows a very strict "Logic Flow."

LONGEST PREFIX MATCH (LPM) – If a destination IP matches two different lines in the routing table, the router chooses the **most specific** one (the one with the longest subnet mask).

- **Example:**
 - Route A: 192.168.1.0 /24
 - Route B: 192.168.1.0 /28
 - If a packet is going to 192.168.1.5, it matches both! But it chooses **Route B** because /28 is more specific than /24.

ADMINISTRATIVE DISTANCE (AD) – The "Trustworthiness" scale of a route source. Lower is better.

- **The "Trust" Rankings:**
 1. **Directly Connected (AD 0):** "I can see it myself." (Most Trusted)
 2. **Static Route (AD 1):** "My boss told me this is where it is."
 3. **OSPF (AD 110):** "My trusted colleague told me."
 4. **Unknown/Untrusted (AD 255):** "I don't believe this route." (Data is dropped)

References:

- Kurose, J. F., & Ross, K. W. (2021). *Computer Networking: A Top-Down Approach*. Pearson.